

Geospatial Data from a 2007 Journey with a Llama Caravan

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Report: <https://escholarship.org/uc/item/1vt6w05j>

Repository: <https://github.com/ntrip6/caravan2007>

Data hosted on Zenodo

Introduction

These data derive from a project I directed in southern Peru in July 2007.

Further information can be gathered from publications (Tripcevich 2010, 2016) and a [video posted online \(Tripcevich 2008\)](#).

Field Setting

In July 2007 Fidel Cruz and his cousin Tadeo Ancco repeated a traditional journey with 28 llamas, their two younger male relatives. They allowed us (four investigators from Peru and the United States) to travel with them as well as Virginia Gutiérrez as they bought salt from villagers who manage an ancient salt quarry, that they then transported to their home base at Chancara (Arequipa). From there, the four male caravaneros prepared for a journey of approximately 90 km northward to a river valley area that lacks local salt deposits. In the northern valley they traded salt for maize, tubers, and other goods, and returned to their home at Chancara. From there we concluded our journey by walking out to Pampamarca.

Spatial Data Gathering Methods

The study goals were focused on travel speeds as a function of slope (testing the applicability of Tobler's hiking function), and documenting how the caravan managed obstacles like river crossings, steep and narrow trail, dense settlements with agricultural fields, and encounters with other caravans. Each morning prior to departure a GPS datalogger was secured to the back of the focal llama, *Cantu* and set to a 2 or 3 second logging interval. The data had to be downloaded to a laptop every night as the logger would fill up. Due to cold weather battery failure data was intermittent on July 21 (see the simplified route layer for the actual route).

Analysis

GPS iTrek data logger provided relatively clean data for a non-differential GNSS. There is no SBAS (WAAS) correction in South America in 2007. The instrument was tested in the city of Arequipa prior to the caravan (Figure 1). From the Caravan data all positional data should be included where slope < 50 ° and speed is < 30 km/hr, data exceeding these values are errant.

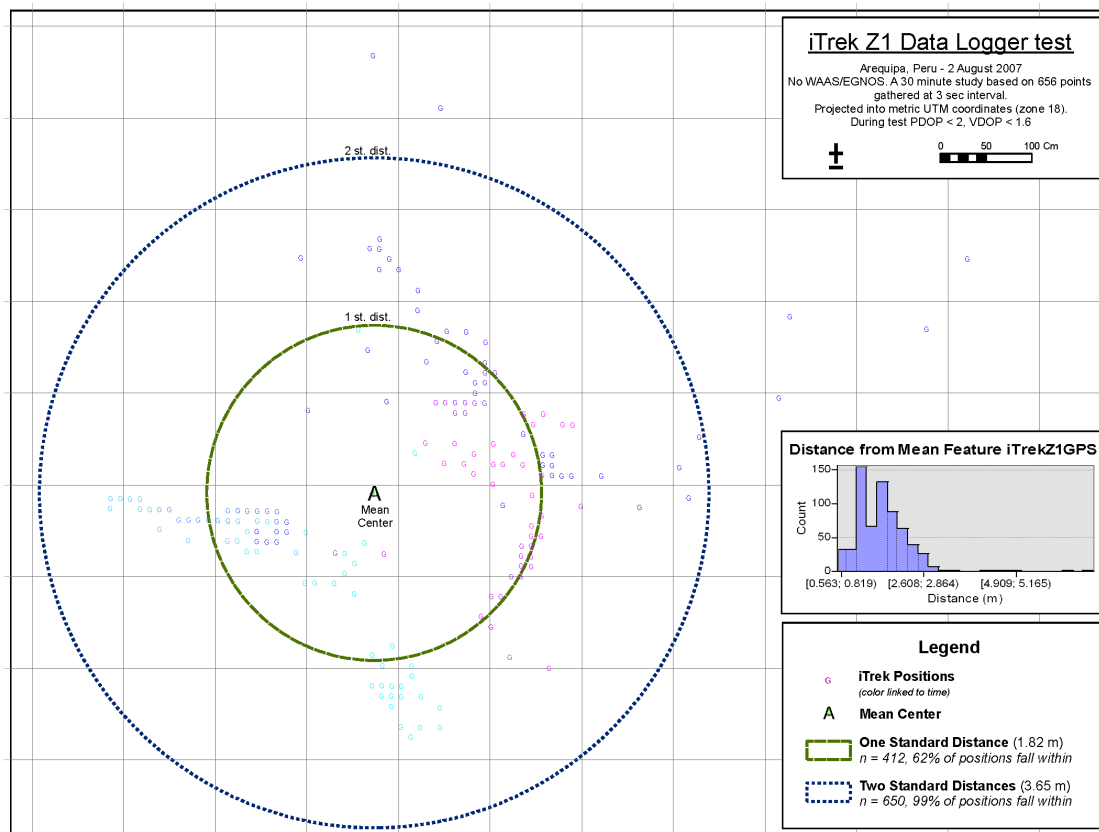


Figure 1. Stationary Standard Distance testing of the iTrek Z1 Data Logger.

Datasets

Caravan Camps KML: Points showing campsite locations, name, and dates of stay.

Caravan Simplified Route KML: a generalized line showing our route.

Caravan_GNSS_Points.csv: a table with all GNSS readings containing the following fields

IDINDEX: Day of month + incrementing daily ID number

LATITUDE: decimal degrees

LONGITUDE: decimal degrees

ELEVATION: Meters above sea level

SECONDS: 2 or 3 second interval until the next GNSS position

TRAILRANK: Ranked assessment of trail quality due to smoothness or width, independent of slope. Values: 0 or NULL no data, 1-2 fast travel, 3-5 slow travel.

Caravan2007_UTM18South.gpkg: A Geopackage containing these GNSS data projected into the local metric coordinate system (UTM Zone 18 south) together with the overnight camp locations and a simplified route.

Caravan_QGIS_Project.qgz: A QGIS3 project file displaying these layers.

Participants

Llamereros

- Fidel Cruz Anco, Virginia Gutiérrez, Rómulo Cruz
- Tadeo Ancco, Raúl Ancco

Investigators

- Nicholas Tripcevich, Félix Palacios, Edison Mendoza, Cheyla Samuelson, Willy Yepez (Arequipa)

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References

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